Semester 7													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24INT451T	Summer Internship	0	0	0	0	2	--	100	--	--	--	100
OE-4	Open Elective 4 (Inter-department, FoET)	3	0	0	3	3	25	25	50	--	--	100
24CH401T	Process Modelling and Optimization	3	0	0	3	3	25	25	50	--	--	100
24CH401P	Process Modelling and Optimization Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH402T	Transport Phenomena	3	1	0	4	4	25	25	50	--	--	100
24CH403T	Computer Aided Process Design	3	0	0	3	3	25	25	50	--	--	100
24CH403P	Computer Aided Process Design Laboratory	0	0	2	2	1	--	--	--	50	50	100
PE-4	Programme Elective 4	Students must choose one course from the given basket of elective courses for study										
24CH431T	Polymer Science & Technology	3	0	0	3	3	25	25	50	--	--	100
24CH432T	Energy Conversion Device Engineering											
24CH433T	Pharmaceuticals Technology											
PE-5	Programme Elective 5	Students must choose one course from the given basket of elective courses for study										
24CH434T	Process Plant Safety, Health and Hygiene	3	0	0	3	3	25	25	50	--	--	100
24CH435T	Project Management											
24CH436T	Plant Design & Process Economics											
24PRCH451T	Seminar	0	0	0	0	0	--	--	--	--	--	Audit
	Total	**18**	**1**	**6**	**23**	**23**						

Semester 8													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24PRCH452T/ 24PRCH453T	Major Project / Comprehensive Project	0	0	26	26	13	--	100	--	--	--	100
	Total	**0**	**0**	**26**	**26**	**13**						

Note: Students shall opt for any one of 24YOG101 / 24NSS101 / 24NCC101.
Note: 24INT251T Industrial Orientation is a mandatory audit (PP/NP) component carrying zero credits.
Note: Students shall register for any one of 24PRCH452T Major Project or 24PRCH453T Comprehensive Project.
Total credits for the award of the degree: 160

24INT451T					Summer Internship					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	0	2	0	--	--	100 (CE)	--	--	100

COURSE OBJECTIVES

- To expose students to the professional working environment of an industry or research organization.
- To relate the theoretical knowledge of chemical engineering to industrial practice.
- To develop the ability to work on a defined problem under professional supervision.
- To develop technical documentation, presentation and professional communication skills.

COURSE CONTENT / ACTIVITIES

1. Identification of the host organization and the problem statement in consultation with the industry supervisor and faculty mentor.
2. Familiarization with the organization, its processes, safety practices and standard operating procedures.
3. Execution of the assigned work - process study, data collection, analysis, design, simulation or experimentation as applicable.
4. Maintenance of a daily logbook duly certified by the industry supervisor.
5. Submission of the internship report and presentation with viva-voce before the departmental evaluation committee.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall the organizational structure and process operations of the host organization.
CO2 - Understand the industrial practices, standards and safety norms followed at the workplace.
CO3 - Apply chemical engineering fundamentals to the problem assigned during the internship.
CO4 - Analyse plant or laboratory data and interpret the results meaningfully.
CO5 - Evaluate the outcome of the work carried out and suggest improvements.
CO6 - Communicate the work through a professional report and presentation.

TEXT / REFERENCE BOOKS

1. Pandit Deendayal Energy University. (2024). *Summer internship guidelines and logbook*. PDEU.
2. Sinnott, R. K., & Towler, G. (2020). *Chemical engineering design* (6th ed.). Butterworth-Heinemann.

EVALUATION

Evaluation is by Continuous Evaluation (CE) for 100 marks, comprising the industry supervisor's assessment, logbook, internship report, presentation and viva-voce.

24CH401T					Process Modelling and Optimization					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To make students understand the various essential features of physical and mathematical modelling and their applications to chemical engineering processes.
- To develop mathematical models and obtain solutions for these models using analytical and numerical methods, and to use commercial process simulators.
- To provide an overview of state-of-the-art optimization algorithms and the theoretical principles that support them, and to provide the modelling skills necessary to describe and formulate optimization problems.
- To develop the technical skills for solving several types of practically relevant optimization problems arising in process systems engineering.

Unit I: OVERVIEW OF MODELLING OF CHEMICAL PROCESSES **Hours: 9**

Introduction to mathematical modelling: uses of mathematical models, scope of coverage, principles of formulation. Fundamental laws: continuity equations, energy equations, equations of motion, transport equations, equations of state, equilibrium and chemical kinetics.

Unit II: MATHEMATICAL MODEL DEVELOPMENT **Hours: 9**

Examples of mathematical models of chemical engineering systems: series of isothermal constant-holdup CSTRs; CSTRs with variable holdups; two heated tanks; gas phase pressurised CSTRs; non-isothermal CSTR; single component vaporizer; multi-component flash drum; batch reactor; reactor with mass transfer.

Unit III: INTRODUCTION TO OPTIMIZATION AND TRADITIONAL OPTIMIZATION TECHNIQUES **Hours: 14**

Single variable optimization algorithms: optimal problem formulation, optimization algorithms, optimality criteria, bracketing methods, region elimination methods, point-estimation method, gradient based methods, root finding using optimization techniques. Multivariable optimization algorithms: unidirectional search, direct search methods, linearized search techniques, generalized reduced gradient method, simplex method.

Unit IV: NON-TRADITIONAL OPTIMIZATION TECHNIQUES **Hours: 7**

Specialized algorithms: integer programming. Non-traditional optimization algorithms: genetic algorithms, simulated annealing, optimization using artificial neural networks.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Relate the important physical phenomena from the problem statement and identify various types of models such as empirical and hybrid models.
CO2 - Translate chemical process systems into mass balance, energy balance and momentum balance formulations.
CO3 - Develop model equations for a given system and demonstrate the model-solving ability for various processes and unit operations such as reactors, distillation columns and heat exchangers, and analyse their behaviour.
CO4 - Theoretically examine different types of optimization problems.
CO5 - Assess the utilization of different optimization techniques.
CO6 - Solve various multivariable optimization problems.

TEXT / REFERENCE BOOKS

1. Edgar, T. F., Himmelblau, D. M., & Lasdon, L. S. (2001). *Optimization of chemical processes* (2nd ed.). McGraw Hill.
2. Luyben, W. L. (1996). *Process modeling, simulation and control for chemical engineers* (2nd ed.). McGraw-Hill.
3. Bequette, B. W. (1998). *Process dynamics: Modeling, analysis and simulation*. Prentice Hall International.
4. Deb, K. (2012). *Optimization for engineering design: Algorithms and examples* (2nd ed.). PHI Learning.
5. Babu, B. V. (2007). *Process plant simulation*. Oxford Higher Education.
6. Sharma, J. K. (2016). *Operations research: Theory and applications* (5th ed.). Macmillan.

24CH401P					Process Modelling and Optimization Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To acquire a foundational understanding of computational tools for computer-aided calculations and the utilization of specialized toolboxes.
- To develop skills in coding and model development for regression analysis, optimization problems and solving ordinary differential equations using computational software.
- To learn the application of computational techniques for analyzing and solving complex engineering problems, including both linear and non-linear scenarios.
- To enhance problem-solving capabilities through practical exercises in coding, model development and the application of optimization algorithms in engineering contexts.

LIST OF EXPERIMENTS

1. Reacquaintance with the GUI of computational software and basic computation (open source and proprietary).
2. First and second order numerical differentiation of functions using forward, backward and central finite difference expressions.
3. Curve fitting (using coding as well as toolboxes) and non-linear and polynomial regression analysis of chemical engineering data for model development.
4. Develop and solve chemical engineering models (first order ODEs) using in-built functions in computational tools.
5. Develop and solve chemical engineering models (first order ODEs) using ODE solving algorithms such as Runge-Kutta 1, 2 and 4.
6. Develop and solve chemical engineering models (higher order ODEs) using in-built functions in computational tools.
7. Optimization algorithm coding in computational tools (open methods such as secant and Newton-Raphson).
8. Optimization algorithm coding in computational tools (bracketed methods such as bisection and regula-falsi).
9. Solving chemical engineering problems in optimization: linear and mixed integer programming.
10. Practising optimization problems using sophisticated algorithms: Box's evolutionary algorithm.
11. Practising optimization problems using sophisticated algorithms: simplex search method.
12. Basic understanding and application of the ANN toolbox.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the GUIs of computational software for chemical engineering applications.
CO2 - Apply numerical differentiation and curve fitting to chemical engineering data analysis.
CO3 - Analyse and solve first-order ODE models using computational tools.
CO4 - Solve higher-order ODEs in chemical engineering using computational methods.
CO5 - Design optimization algorithms for chemical engineering problems.
CO6 - Apply specialized toolboxes for chemical engineering solutions.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Process modelling and optimization laboratory manual*. Pandit Deendayal Energy University.
2. Chapra, S. C., & Canale, R. P. (2021). *Numerical methods for engineers* (8th ed.). McGraw-Hill Education.
3. Law, V. J. (2013). *Numerical methods for chemical engineers using Excel, VBA, and MATLAB*. CRC Press.
4. Kharab, A., & Guenther, R. B. (2023). *An introduction to numerical methods: A MATLAB approach* (5th ed.). CRC Press.
5. Gilat, A. (2007). *Numerical methods for engineers and scientists: An introduction with applications using MATLAB*. Wiley.
6. King, M. R., & Mody, N. A. (2010). *Numerical and statistical methods for bioengineering: Applications in MATLAB*. Cambridge University Press.

24CH402T					Transport Phenomena					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

- To identify the transport processes involved in some simple situations.
- To understand the equations of change for mass, momentum and energy and the requisite boundary conditions for each equation of change.
- To solve the simplified problem analytically with the requisite mathematical framework and assess the physical significance of the solution obtained.
- To examine the range of applicability and the limitations of the solution.

Unit I: FUNDAMENTALS OF TRANSPORT PROCESSES **Hours: 12**

Vector and tensor analysis; curvilinear coordinates; continuum approximation; Eulerian and Lagrangian approaches; Reynolds transport theorem; general forms of conservation equations; conservation of mass, momentum and energy; constitutive equations; viscosity and the mechanism of momentum transport; thermal conductivity and the mechanism of energy transport; diffusivity and the mechanism of mass transport.

Unit II: SHELL BALANCE METHOD **Hours: 12**

Application of shell momentum balance to Cartesian, cylindrical and spherical coordinate systems; derivation of momentum transport, heat transport and mass transport equations using steady state shell balances in different coordinate systems; different types of boundary conditions; solving simple transport problems using shell balances.

Unit III: EQUATIONS OF CHANGE **Hours: 16**

Application of frame-independent forms of momentum, energy and mass conservation equations; dimensional analysis and different flow regimes; obtaining analytical solutions of transport problems using the Navier-Stokes and Cauchy momentum equations and energy and mass transport equations in different coordinate systems; coupling between various equations and a brief introduction to CFD.

Unit IV: MORE THAN ONE INDEPENDENT VARIABLE SYSTEMS **Hours: 12**

Solving transport problems with more than one independent variable: velocity distribution, temperature distribution and concentration distribution; CFD-based transport modelling (one demonstration example); introduction to turbulent flow.

Max 52 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the momentum, energy and mass fluxes and their individual components.
CO2 - Compare various coordinate systems and accordingly identify the direction of forces and fluxes on a given type of shell.
CO3 - Apply shell balance methods to obtain equations of momentum, energy and mass transport in terms of flux quantities.
CO4 - Simplify various transport problems by using various equations of change.
CO5 - Choose boundary conditions and mathematical techniques to solve equations of change.
CO6 - Compose and formulate moderately complex transport problems from scratch.

TEXT / REFERENCE BOOKS

1. Bird, R. B., Stewart, W. E., & Lightfoot, E. N. (2006). *Transport phenomena* (2nd ed.). John Wiley & Sons.
2. Leal, L. G. (2007). *Advanced transport phenomena: Fluid mechanics and convective transport processes*. Cambridge University Press.
3. Deen, W. M. (1998). *Analysis of transport phenomena*. Oxford University Press.

24CH403T					Computer Aided Process Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the scope of process and plant design using software tools and to create, select and apply appropriate modern software tools, including modelling, to complex chemical engineering processes with an understanding of the limitations.
- To identify the components of physical and thermodynamic property models and learn the software aspects of rapid solution.
- To learn applications and solve chemical process flowsheeting problems more quickly, efficiently and successfully using computer-aided tools.
- To learn the concepts of energy integration and solve heat exchanger network problems in an energy analyser.

Unit I: BASICS OF COMPUTER AIDED PROCESS DESIGN **Hours: 12**

Integrated process plant design with computer-aided tools; PFD, P&ID, pre-FEED and FEED stages; process simulators - components and architecture, blocks and streams library; CAPD problem formulation and solvers; model equations and degrees of freedom; design, rating, specification and optimization formulation; sequential modular and equation oriented approaches; tearing of streams; scope of Aspen Plus/HYSYS simulators; flowsheeting problem solving using design specifications and sensitivity analysis; incorporating the cost of safety systems in the evaluation of process systems; plant layout with safety aspects in mind.

Unit II: COMPUTERISED PHYSICAL AND THERMODYNAMIC PROPERTY METHODS AND MODELS **Hours: 9**

Computerized physical and thermodynamic properties, methods and models; ideal and non-ideal model selection criteria; thermodynamic databanks; NIST data bank; estimation of activity coefficient and EOS model parameters using data regression systems; case studies and flowsheeting problem solving of mass and energy balance, phase equilibrium, chemical equilibrium and reactor design using commercial software; hazardous area classification using process simulation and thermodynamic insight.

Unit III: COMPUTER SIMULATION OF MASS AND HEAT TRANSFER EQUIPMENT **Hours: 9**

Design and simulation of distillation and absorption columns - equilibrium and rate-based approaches; design, rating and simulation of heat exchangers; petroleum fractionation columns; various unit models available in the simulator library and their usage.

Unit IV: PROCESS INTEGRATION AND HEAT EXCHANGER NETWORKS **Hours: 9**

Introduction to process intensification and process integration; heat exchanger networks (HEN); cold, hot composite and utility curves; pinch point analysis; solving HEN problems using an energy analyser.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand, select and describe the fundamentals and computer tools for chemical industrial process simulation applications.
CO2 - Understand, choose and apply the unit operation/process blocks and material/energy streams from a simulator library to make a process flowsheet.
CO3 - Identify, choose and justify the components of physical and thermodynamic property methods and models and their usage in software aspects of rapid solution.
CO4 - Determine flowsheeting solutions and design calculations by using design specification and sensitivity analysis approaches.
CO5 - Solve process design and simulation calculations of various unit operations and processes using process simulation software.
CO6 - Outline energy integration and construct a heat exchanger network using the composite curve approach.

TEXT / REFERENCE BOOKS

1. Haydary, J. (2019). *Chemical process design and simulation*. John Wiley & Sons.
2. Finlayson, B. A. (2014). *Introduction to chemical engineering computing* (2nd ed.). Wiley Interscience.
3. Jana, A. K. (2012). *Process simulation and control using Aspen* (2nd ed.). PHI Learning.
4. Al-Malah, K. I. M. (2016). *Aspen Plus: Chemical engineering applications*. John Wiley & Sons.

24CH403P					Computer Aided Process Design Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To understand the software to create, select and apply appropriate modelling and simulation of complex chemical engineering processes with an understanding of the limitations.
- To identify and apply the components of physical and thermodynamic property methods and models and learn the software aspects of rapid solution.
- To learn applications and solve chemical process flowsheeting problems more quickly, efficiently and successfully using computer-aided tools.
- To learn the concepts of energy integration and solve heat exchanger network problems.

LIST OF EXPERIMENTS

1. To construct a process flow diagram and simulation sheet with process blocks and streams.
2. To estimate physical properties such as critical properties, ΔH and ΔG of organic compounds.
3. Thermodynamic property estimation and analysis: T-xy, P-xy and x-y diagrams of a binary mixture.
4. Regression of vapour-liquid equilibrium data; flash separation; dew point and bubble point calculations.
5. Gibbs reactor, equilibrium reactor and yield reactor.
6. Kinetic reactors: batch, CSTR and PFR.
7. Heat exchangers: design, rating and calculation with TEMA specifications.
8. Thermal analysis and simulation of a heat exchanger.
9. Design and simulation of distillation and absorption columns.
10. Modelling of azeotropic distillation using pressure swing with RadFrac.
11. Case study: ammonia production process - open loop process.
12. Costing and economic analysis; case study: ammonia production process - closed loop process.
13. Solving a heat exchanger network using an energy analyser.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand and estimate the physical and thermodynamic properties of organic compounds.
CO2 - Develop a flowsheet for mass and energy balance calculations using process simulation software.
CO3 - Construct thermodynamic phase diagrams (T-xy, P-xy, x-y) using activity coefficient and equation of state models, and analyse and compare them.
CO4 - Determine design solutions by using design specification and sensitivity analysis approaches.
CO5 - Verify and compare process design and simulation calculations of various unit operations.
CO6 - Demonstrate heat exchanger network design and composite curves using an energy analyser.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Computer aided process design laboratory manual*. Pandit Deendayal Energy University.
2. Haydary, J. (2019). *Chemical process design and simulation*. John Wiley & Sons.
3. Finlayson, B. A. (2014). *Introduction to chemical engineering computing* (2nd ed.). Wiley Interscience.
4. Jana, A. K. (2012). *Process simulation and control using Aspen* (2nd ed.). PHI Learning.
5. Al-Malah, K. I. M. (2016). *Aspen Plus: Chemical engineering applications*. John Wiley & Sons.

24CH431T					Polymer Science & Technology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 4)

COURSE OBJECTIVES

- To prepare students with basic knowledge of polymer science and technology that will help them develop new materials.
- To impart critical knowledge about various complexities in polymerization reaction engineering.
- To teach students various characterization techniques for identification, molecular weight calculation, etc.
- To prepare students for mastering a few bulk polymer process technologies.

Unit I: INTRODUCTION TO POLYMER SCIENCE **Hours: 9**

Classification and nomenclature of polymers; monomer and functionality; polymerization and degree of polymerization; types of molecular weights and their distribution; molecular weight determination; factors affecting molecular weight and molecular weight distribution; chain-transfer reactions; thermal and mechanical properties of polymers; concept of stereochemistry of polymers.

Unit II: CHEMISTRY OF POLYMERIZATION **Hours: 9**

Classification of polymerization mechanisms; stepwise polymerization; kinetics and statistics of linear stepwise polymerization; radical chain (addition) polymerization - reaction mechanisms and associated kinetics; copolymerization - reaction mechanisms and kinetics; techniques of polymerization: bulk, solution, suspension and emulsion.

Unit III: INDUSTRIAL POLYMER PRODUCTION AND SPECIALITY POLYMERS **Hours: 12**

Production of bulk polymers: polyethylene, polypropylene, polyvinylchloride, styrene butadiene rubber, polyester and polyamide (Nylon); speciality polymers: conducting polymers, block copolymers, polymer composites, polymer blends, polyurethanes, silicones and other inorganic polymers - rubbers, epichlorohydrin, polysulphides, acrylic rubbers and silane-containing polymers.

Unit IV: POLYMER PROCESSING TECHNIQUES AND ENTREPRENEURSHIP IN POLYMERS **Hours: 9**

Polymer processing: injection moulding, extrusion, thermoforming, calendaring, compression moulding, blow moulding, spinning of fibres, vulcanization and reinforcement of rubbers; importance of Tg in processing; dependence of small and medium scale industries on polymer processing; entrepreneurship opportunities available in polymer processing.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - List and recall the basic concepts of polymer science.
CO2 - Classify and compare different polymerization reactions and interpret polymerization methods judiciously.
CO3 - Identify and develop the knowledge of different processes of bulk polymer production.
CO4 - Discover the importance of speciality polymers for different industrial and day-to-day applications.
CO5 - Assess and compare the different techniques of polymer processing in industries.
CO6 - Identify the various entrepreneurship opportunities available in polymer processing.

TEXT / REFERENCE BOOKS

1. Billmeyer, F. W. (2008). *Textbook of polymer science* (3rd ed.). John Wiley & Sons.
2. Gowariker, V. R., Viswanathan, N. V., & Sreedhar, J. (2016). *Polymer science*. New Age International.
3. Stevens, M. P. (1999). *Polymer chemistry: An introduction* (3rd ed.). Oxford University Press.
4. Odian, G. (2004). *Principles of polymerization* (4th ed.). Wiley.

24CH432T					Energy Conversion Device Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 4)

COURSE OBJECTIVES

- To know the basic scientific processes behind various energy conversion devices, namely batteries, fuel cells, photo(electro)chemical hydrogen generators and solar cells.
- To understand the fabrication, working and disposal of these energy conversion devices.
- To build engineering perspectives on efficient energy generation using these devices.
- To design methods for networking, stacking and multiplexing the devices for optimum usage in process plants.

Unit I: INTRODUCTION AND BASIC PRINCIPLES **Hours: 12**

Electrochemical cells; characteristics of electrochemical reactions; importance of electrochemical systems; Faraday's law; cell potentials; Nernst equation and variants; standard potentials; use of the cell potential; Pourbaix diagrams; reference electrodes; impact of potential on reaction rate; Butler-Volmer equation; the influence of mass transfer on the reaction rate; kinetics in full cells; various efficiency measurements in electrochemical systems; electric and hybrid systems; power demand in vehicles; hybrid vehicle architectures.

Unit II: BATTERY FUNDAMENTALS **Hours: 9**

Components of a cell; classification of batteries and cell chemistries; theoretical capacity and the state of charge; cell characteristics and electrochemical performance; Ragone plots; efficiency of secondary cells; charge retention and self-discharge; capacity fade in secondary cells; redox-flow batteries; scaling of cells to adjust capacity; thermal management; mechanical considerations; battery electric vehicles; batteries for full-hybrid electric vehicles.

Unit III: FUEL-CELL FUNDAMENTALS **Hours: 9**

Types and components of fuel cells; current-voltage characteristics and polarizations; effect of operating conditions and maximum power; electrode structure; proton-exchange membrane fuel cells; solid oxide fuel cells; basic stack design concepts; cell stack configurations; utilization of oxidant and fuel; flow-field design; water and thermal management; structural-mechanical considerations; fuel-cell hybrid systems for vehicles.

Unit IV: PHOTOELECTROCHEMICAL CELLS AND ELECTROCHEMICAL DOUBLE-LAYER CAPACITORS **Hours: 9**

Semiconductor basics; energy scales; semiconductor-electrolyte interface; current flow in the dark; light absorption; photoelectrochemical effects; photoelectrochemical cells; electrical double-layer capacitance; current-voltage relationship for capacitors; electrode structure; impedance analysis; full cell analysis; power and energy capabilities; cell design; practical operation and electrochemical capacitor performance; pseudo-capacitance; applications in hybrid and electric vehicles.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the basics of electrochemical energy conversion and its use in vehicular applications.
CO2 - Demonstrate the testing procedures and characterisation of energy conversion devices.
CO3 - Apply the energy conversion principles to augment the engineering of the device.
CO4 - Evaluate the performance of energy conversion devices in vehicles based on laboratory tests.
CO5 - Analyse practical application challenges in energy conversion devices.
CO6 - Design a networked system of energy conversion devices to meet vehicular power demand.

TEXT / REFERENCE BOOKS

1. Fuller, T. F., & Harb, J. N. (2018). *Electrochemical engineering*. Wiley.
2. Lvov, S. N. (2015). *Electrochemical science and engineering*. CRC Press.
3. Alkire, R. C., Bartlett, P. N., & Koper, M. T. (Eds.). (2019). *Advances in electrochemical science and engineering: Vol. 18. Electrochemical engineering - The path from discovery to product*. Wiley-VCH.
4. Bard, A. J., Stratmann, M., Macdonald, D. D., & Schmuki, P. (2007). *Encyclopaedia of electrochemistry: Vol. 5. Electrochemical engineering*. Wiley-VCH.

24CH433T					Pharmaceuticals Technology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 4)

COURSE OBJECTIVES

- To acquire the knowledge of basic pharmaceutical technology and characteristics of pharmaceutical industries.
- To gain an understanding of sterilization applications and dosage forms.
- To understand and apply methods for the manufacturing and packaging of pharmaceutical products.
- To understand advanced pharmaceutical manufacturing and the use of AI in the pharmaceutical industry.

Unit I: BASICS OF PHARMACEUTICAL TECHNOLOGY **Hours: 8**

Characteristics of pharmaceutical industries; product standards - IP, BP, USP; methods of production; AI in pharmaceutical production; chemical synthesis; isolation from plants; isolation from animals; fermentation; API and formulation; nanomedicine; drug design for targeted drug delivery; diffusion modelling of drug delivery and coding to solve.

Unit II: STERILIZATION AND CHARACTERIZATION **Hours: 10**

Sterility and the requirement of sterility; concept of bacterial and viral infection; natural growth of bacteria; fundamental concepts of bacterial culture; concept of sterilization; methods of sterilization with applications - heat sterilization, steam sterilization, dry heat sterilization, radiation sterilization, gas sterilization and filtration sterilization; sterile facilities; use of UV-Vis, FTIR, Raman spectroscopy and HPLC in the pharmaceutical industry.

Unit III: CLEAN FACILITIES, CLEAN ROOM AND DOSAGE FORMS **Hours: 11**

Design parameters for clean facilities: air change rate, pressurization, temperature control and humidity control; architectural design issues; materials of construction for walls, doors, ceilings and floors; clean construction; HEPA filters and their production by electrospinning; air quality monitoring; indoor air quality (IAQ); HVAC unit; clean room facility; role of artificial intelligence in air quality monitoring. Solid dosage forms: tablets, coated tablets, gelatine capsules, chewable tablets and excipients. Semi-solid dosage forms: ointments and creams, bases for ointments and creams, types of gels and commercial gelling agents.

Unit IV: MANUFACTURING AND PACKAGING AND USE OF AI **Hours: 10**

Manufacturing of tablets and capsules; advanced manufacturing in micro-reactors; usage of AI in advanced drug manufacturing units; packaging and storage of ointments and creams; critical aspects of liquid manufacturing - particle size of raw materials, parameters of compounding, uniformity and stability problems; packaging materials - general considerations, glass, plastic and metal; quality control of packaging materials.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the key concepts of pharmaceutical technology.
CO2 - Identify appropriate methods in medicine production.
CO3 - Apply various methods of sterilization.
CO4 - Explain air filters, clean rooms, HVAC units and AI-based indoor air quality monitoring.
CO5 - Produce different dosage forms and select packaging materials for pharmaceutical products.
CO6 - Discuss the role of AI in pharmaceutical industries.

TEXT / REFERENCE BOOKS

1. Hickey, A. J., & Ganderton, D. (2001). *Pharmaceutical process engineering*. Marcel Dekker.
2. Gad, S. C. (2008). *Pharmaceutical manufacturing handbook: Production and processes*. John Wiley & Sons.
3. Sharp, J. (2005). *Good pharmaceutical manufacturing practice: Rationale and compliance*. CRC Press.
4. Aulton, M. E., & Taylor, K. M. G. (2017). *Aulton's pharmaceutics: The design and manufacture of medicines* (5th ed.). Elsevier.

24CH434T					Process Plant Safety, Health and Hygiene					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 5)

COURSE OBJECTIVES

- To learn about engineering ethics, accident and loss statistics and hazard identification.
- To have knowledge of hazard analysis, toxicology, personal protective equipment and risk assessment.
- To learn about fire and explosion hazards, their prevention and risk analysis.
- To gain the knowledge of identification, evaluation and control of industrial hygiene.

Unit I: INTRODUCTION **Hours: 12**

Engineering ethics; accident and loss statistics; nature of the accident process; inherent safety in process; indoor air quality management; personal safety aspects; PPE; toxicology; national and international standards for health and safety; acts and legislation (OSHA, ISO standards); REACH regulations. Hazard identification: hazard surveys, hazard and operability study, hazard analysis.

Unit II: FIRE AND EXPLOSION HAZARDS **Hours: 9**

Fire triangle; flammability characteristics of fluids; limiting oxygen concentration; flammability diagram; explosions; DOW fire and explosion index; prevention - inerting, ventilation, sprinkler systems, explosion proof instruments, equipment and control rooms; miscellaneous concepts for preventing fires and explosions.

Unit III: INDUSTRIAL HYGIENE **Hours: 9**

Government regulations; identification - material safety data sheet; evaluation and control. Risk assessment: fault tree analysis (FTA), event tree analysis (ETA), review of probability theory, quantitative risk analysis, LOPA concept.

Unit IV: MAJOR DISASTERS AND MISCELLANEOUS CONCEPTS **Hours: 9**

Case studies on major disasters in the chemical process industry; introduction to relief and sizing of relief systems; location of flare systems and knockout pots.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define engineering ethics, safety principles, toxicology, hazards and industrial hygiene.
CO2 - Classify protective measures to mitigate risk and implement safety, health and industrial hygiene.
CO3 - Apply knowledge in risk management and accident and loss statistics.
CO4 - Examine industrial hygiene, risk assessment, probability theory, quantitative risk analysis and the LOPA concept.
CO5 - Assess typical sources of risk in a process plant by hazard identification and case studies.
CO6 - Estimate fire and explosion hazards and their identification, evaluation and control.

TEXT / REFERENCE BOOKS

1. Crowl, D. A., & Louvar, J. F. (2011). *Chemical process safety: Fundamentals with applications* (3rd ed.). Prentice Hall.
2. Cheremisinoff, N. P. (2001). *Practical guide to industrial safety: Methods for process safety professionals*. Marcel Dekker.
3. Wilson, L., McCutcheon, D., & Buchanan, M. (2003). *Industrial safety and risk management*. University of Alberta Press.
4. Center for Chemical Process Safety. (2008). *Guidelines for hazard evaluation procedures* (3rd ed.). AIChE-Wiley.

24CH435T					Project Management					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 5)

COURSE OBJECTIVES

- To acquire and fine-tune the skills and techniques for the four phases in the life cycle of a typical project.
- To gain an understanding of the essential principles associated with effective project management.
- To understand and apply methods for solving common difficulties associated with project management.
- To gain knowledge of AI/ML based project management concepts.

Unit I: INTRODUCTION TO PROJECT MANAGEMENT **Hours: 9**

Definition of a project; project management vs. general management; the life cycle of projects; confronting uncertainty; project portfolio process; an approach to project formulation; the project manager's roles; the project manager's responsibility to the project; selection of a project manager; fitting projects into the parent organization; the project team.

Unit II: PLANNING AND BUDGETING THE PROJECT **Hours: 9**

The contents of a project plan; the planning process - detailed overview; the work breakdown structure and other aids; multidisciplinary teams - balancing pleasure and pain; methods of budgeting; cost estimating; improving cost estimates; budget uncertainty and risk management.

Unit III: SCHEDULING AND ALLOCATING THE RESOURCES TO THE PROJECT **Hours: 9**

PERT and CPM networks; project uncertainty and risk management; simulation; machine learning based risk assessment and project management; the Gantt chart; extensions to PERT/CPM; expediting a project; resource loading; resource levelling; allocating scarce resources to projects.

Unit IV: MONITORING AND CONTROLLING THE PROJECT **Hours: 12**

Project monitoring and control: project work measurement, performance measurement, earned value management, estimate revision, control of overrun of a project in time and cost. Project closure and review: performance evaluation - scope, time and cost; performance of teams; lessons learnt; project closure report; AI-based cost analysis, performance monitoring and preparation of the closure report.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the key concepts of project management and the project lifecycle.
CO2 - Analyse the principles of management, planning, specifications and requirements.
CO3 - Understand the processes involved in the effective planning and subsequent management of projects.
CO4 - Construct the key stages of managing projects, scheduling of tasks and risk analysis, and describe the role of the project manager through techniques such as CPM, PERT and Gantt charts.
CO5 - Develop increased awareness of available resources to further develop project management skills.
CO6 - Elaborate new knowledge for their own projects, set realistic goals for moving forward and appreciate the role of AI in project management.

TEXT / REFERENCE BOOKS

1. Maylor, H. (2010). *Project management* (4th ed.). Pearson Education.
2. Pinto, J. K. (2009). *Project management: Achieving competitive advantage*. Pearson Education.
3. Khatua, S. (2011). *Project management and appraisal*. Oxford University Press.
4. Meredith, J. R., & Mantel, S. J. (2011). *Project management: A managerial approach* (8th ed.). John Wiley & Sons.

24CH436T					Plant Design & Process Economics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 5)

COURSE OBJECTIVES

- To study the basics of process flow diagram formulation.
- To know the detailed-estimate design procedure.
- To understand the effect of design parameters on the process design and the methodology to estimate these design parameters.
- To understand the concepts of economic analysis of the process and the necessity of optimized design.

Unit I: INTRODUCTION TO PLANT DESIGN AND PROCESS ECONOMICS **Hours: 9**

Basic concepts: general design considerations; process design development; layout of plant items; flow sheets and P&I diagrams; economic aspects and optimum design; practical considerations in design and engineering ethics; degrees of freedom analysis in interconnected systems; direct and indirect costs.

Unit II: DESIGN CONSIDERATION AND PROCESS DESIGN DEVELOPMENT **Hours: 9**

Synthesis of flow sheets: propositional logic and semantic equations; deduction theorem; algorithmic flow sheet generation using P-graph theory; sequencing of operating units; feasibility and optimization of flow sheets using algorithms viz. solution structure generation (SSG), maximal structure generation (MSG), simplex and branch-and-bound.

Unit III: ANALYSIS OF COST ESTIMATION FOR PROCESS EQUIPMENT **Hours: 12**

Factors affecting investment and production costs; estimation of capital investment and total product costs; interest; time value of money; break-even analysis; taxes and fixed charges; salvage value; methods of calculating depreciation; profitability; alternative investments and replacements; cost and asset accounting; illustrative case studies in process equipment design and costing of equipment in the categories of material transfer, handling and treatment equipment; heat transfer equipment - shell and tube heat exchangers (Kern and Bell-Delaware design methods), plate heat exchangers, evaporators; mass transfer equipment - absorption/stripping columns (packed/tray), multicomponent distillation column (Fenske-Underwood-Gilliland correlations); reactors - choice of reactors, non-isothermal reactors, reactor configuration, interstage heating/cooling, multi-tubular reactors, catalyst deactivation.

Unit IV: OPTIMIZATION AND DESIGN STRATEGY **Hours: 9**

Optimum production rates in plant operation; optimum batch cycle time applied to evaporators and filter presses; economic pipe diameter; optimum insulation thickness; optimum cooling water flow rate; optimum distillation reflux ratio.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basic concepts of process flow sheet formulation.
CO2 - Evaluate the detailed process design.
CO3 - Know the economic analysis of the overall process.
CO4 - Correlate the design with equipment cost.
CO5 - Find the best optimum conditions used in the design.
CO6 - Analyse the existing process plant performance.

TEXT / REFERENCE BOOKS

1. Peters, M. S., Timmerhaus, K. D., & West, R. E. (2011). *Plant design and economics for chemical engineers* (5th ed.). McGraw Hill.
2. Smith, R. (2016). *Chemical process design and integration* (2nd ed.). Wiley.
3. Towler, G., & Sinnott, R. (2021). *Chemical engineering design: Principles, practice and economics of plant and process design* (3rd ed.). Butterworth-Heinemann.

24PRCH451T					Seminar					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	0	0	0	--	--	--	--	--	Audit (PP/NP)

COURSE OBJECTIVES

- To develop the ability to carry out an independent literature survey on a contemporary topic in chemical engineering.
- To develop technical reading, critical appraisal and synthesis skills.
- To develop skills of technical presentation before a professional audience.
- To prepare students for the major/comprehensive project of Semester 8.

COURSE CONTENT / ACTIVITIES

1. Selection of a contemporary topic in chemical engineering in consultation with the faculty guide.
2. Systematic literature survey using standard databases, journals, patents and standards.
3. Critical appraisal and synthesis of the literature reviewed.
4. Preparation of a seminar report in the prescribed format, checked for plagiarism.
5. Oral presentation followed by question-answer session before the departmental committee.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall the sources of technical literature relevant to chemical engineering.
CO2 - Understand the state of the art in the chosen topic.
CO3 - Apply literature survey techniques to compile relevant information.
CO4 - Analyse and critically appraise the published work in the chosen area.
CO5 - Evaluate research gaps and propose future directions.
CO6 - Communicate technical content effectively through a report and an oral presentation.

TEXT / REFERENCE BOOKS

1. Pandit Deendayal Energy University. (2024). *Seminar and project guidelines*. PDEU.
2. Day, R. A., & Gastel, B. (2016). *How to write and publish a scientific paper* (8th ed.). Cambridge University Press.

EVALUATION

This is a mandatory zero-credit audit course. The award is PP (Passed) / NP (Not Passed) based on the seminar report and the oral presentation evaluated by the departmental committee.